

1. a) $Z \in \{1, 2, 3\}$ $P(Z=1) = .05$ $P(Z=2) = .15$
 $P(Z=3) = .8$

cdf $F(Z) = P(Z \leq z) = \sum_{z_i \leq z} P(Z=z_i)$

check $= U \leq .05, U \leq .2, U \leq 1$

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if  $U \leq .05$ 
  return  $z=1$ 
elseif  $U \leq .2$ 
  return  $z=2$ 
else
  return  $z=3$ 

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b) 2 comparisons. If there are N thresholds the code only needs $N-1$ comparisons excluding the else statement.

c) $X \sim \text{Geometric}(p)$ $F(k) = 1 - (1-p)^k$ $P(X=k) = p(1-p)^{k-1}$
 $k = 1, 2, 3, \dots$

$U \sim \text{Uniform}(0,1)$

$F(k) \geq U$

$F^{-1}(U) = 1 - (1-p)^k = u$

$\Rightarrow 1-u = (1-p)^k$ take natural logarithm of both sides

$\Rightarrow \log(1-u) = \log((1-p)^k)$

$\Rightarrow \log(1-u) = k \log(1-p)$

$\Rightarrow k = \frac{\log(1-u)}{\log(1-p)}$

2. a)

$$f(x) = \frac{1}{\pi(1+x^2)} \quad x \in \mathbb{R}$$

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(y) dy = \int_{-\infty}^x \frac{1}{\pi(1+y^2)} dy$$

$$= \frac{1}{\pi} \int_{-\infty}^x \frac{1}{1+y^2} dy \quad \text{Trig substitution}$$

$$\text{let } y = \tan(\theta)$$

$$x = \tan(\theta) \Rightarrow \theta = \tan^{-1}(x)$$

$$dy = \sec^2(\theta) d\theta$$

$$\text{let } \alpha = -\infty$$

$$\alpha = \tan(\theta_1)$$

$$\Rightarrow \theta_1 = \tan^{-1}(\alpha)$$

$$\lim_{\alpha \rightarrow -\infty} \tan^{-1}(\alpha) = -\frac{\pi}{2}$$

$$\Rightarrow \frac{1}{\pi} \int_{-\frac{\pi}{2}}^{\tan^{-1}(x)} \frac{1}{\tan(\theta)} \cdot \sec^2(\theta) d\theta$$

$$= \frac{1}{\pi} \int_{-\frac{\pi}{2}}^{\tan^{-1}(x)} \frac{\sec^2(\theta)}{\sec(\theta)} d\theta$$

$$= \frac{1}{\pi} \int_{-\frac{\pi}{2}}^{\tan^{-1}(x)} d\theta = \frac{1}{\pi} \left[\theta \right]_{-\frac{\pi}{2}}^{\tan^{-1}(x)} = \frac{1}{\pi} \left(\tan^{-1}(x) + \frac{\pi}{2} \right)$$

$$\boxed{F(x) = \frac{1}{\pi} \left(\tan^{-1}(x) + \frac{\pi}{2} \right)}$$

$$\text{let } U \sim \text{uniform}(0,1)$$

$$F(x) \geq U$$

$$\Rightarrow F^{-1}(u) = \frac{1}{\pi} \left(\tan^{-1}(x) + \frac{\pi}{2} \right) = u$$

$$\Rightarrow \pi u = \tan^{-1}(x) + \frac{\pi}{2}$$

$$\Rightarrow \tan^{-1}(x) = \pi \left(u - \frac{1}{2} \right)$$

$$\Rightarrow x = \tan\left(\pi \left(u - \frac{1}{2} \right)\right)$$

$$\Rightarrow \boxed{F^{-1}(u) = \tan\left(\pi \left(u - \frac{1}{2} \right)\right)}$$

2. b) calculate median by $F(u) = .5$

$$\Rightarrow .5 = \frac{1}{\pi} \int_{-M}^M \frac{1}{(1+x^2)} dx$$

$$.5 = \int_{-M}^M \frac{1}{\pi(1+x^2)} dx$$

let $x = \tan \theta$

$\Rightarrow \frac{1}{\pi} \int_{\tan^{-1}(-M)}^{\tan^{-1}(M)} \sec^2 \theta d\theta$

$\tan \theta_1 = M$

$\tan \theta_2 = -M$

$$\Rightarrow \frac{1}{\pi} \int_{\tan^{-1}(-M)}^{\tan^{-1}(M)} \sec^2 \theta d\theta$$

$$\Rightarrow \frac{1}{\pi} \left[\tan \theta \right]_{\tan^{-1}(-M)}^{\tan^{-1}(M)}$$

$$\Rightarrow \frac{1}{\pi} \left[\tan^{-1}(M) - \tan^{-1}(-M) \right] = \frac{1}{2\pi} \left[\tan^{-1}(M) + \tan^{-1}(M) \right]$$

$$= \frac{1}{\pi} \left[2 \tan^{-1}(M) \right] = \frac{2 \tan^{-1}(M)}{\pi} = \frac{1}{2}$$

$$2 \tan^{-1}(M) = \frac{\pi}{2}$$

$$\tan^{-1}(M) = \frac{\pi}{4}$$

$$M = \tan\left(\frac{\pi}{4}\right) = 1 \Rightarrow \boxed{M=1}$$

$$E[X] = \int_{-\infty}^{\infty} x f(x) dx = \int_{-\infty}^{\infty} \frac{x}{\pi(1+x^2)} dx \quad \text{let } u = 1+x^2$$

$$du = 2x dx$$

$$\Rightarrow \lim_{x \rightarrow \infty} \int_{-\infty}^x \frac{1}{2\pi} \frac{du}{u} = \frac{1}{2\pi} \lim_{x \rightarrow \infty} \left[\ln(x) \right]_{-\infty}^x$$

$$= \frac{1}{2\pi} \lim_{x \rightarrow \infty} \left[\ln(x) - \ln(-x) \right] = DNE$$

$$n=2 \quad \lambda=1$$

3. $X \sim \text{Gamma}(2, 1)$ $f(x) = x e^{-x}$ for $x \geq 0$ (0 otherwise)
 $F(x) = 1 - (1+x)e^{-x}$ (not invertible)
 $g(x) = 1 e^{-1x}$

a) let $\lambda = \frac{1}{2}$

$$\frac{f(x)}{g(x)} \leq c \Rightarrow \frac{x e^{-x}}{\frac{1}{2} e^{-\frac{1}{2}x}} = 2 x e^{-\frac{1}{2}x} \leq c$$

$$\frac{d}{dx} 2 x e^{-\frac{1}{2}x} = 2 e^{-\frac{1}{2}x} - x e^{-\frac{1}{2}x} = 0 \Rightarrow (2-x) e^{-\frac{1}{2}x} = 0$$

$x=2$ is max

@ $x=2$ max occurs

$$\Rightarrow \frac{4 e^{-1}}{c} \leq 1 \Rightarrow \frac{4 e^{-1}}{c} \leq 1 \Rightarrow \frac{4}{e} \cdot \frac{1}{c} \leq 1 \Rightarrow \frac{4}{e} \leq c$$

maximum c value is $\frac{4}{e}$

b) let $U \sim \text{uniform}(0, 1)$

(while-check = 1)

if $U < F(x_i)$

$x = i$

check = 0

else

$i = i + 1$

end $F(x_{i+1}) \cdot F(x_i)$

end

c) Decreasing λ will allow $g(x)$ to better approximate $f(x)$, which will allow for a much better acceptance fraction.

$$4. f(t) = a\lambda_f e^{-\lambda_f t} + (1-a)\lambda_s e^{-\lambda_s t} \quad t \geq 0$$

$$a = 0.9 \quad \lambda_f = 1000 s^{-1} \quad \lambda_s = 10 s^{-1}$$

a) If f is a density $\int_{-\infty}^{\infty} f(t) dt = 1$ $f(t) = 0$ on $-\infty < t < 0$

$$\Rightarrow \int_{-\infty}^{\infty} f(t) dt = \lim_{B \rightarrow \infty} \int_{-B}^B f(t) dt = \lim_{B \rightarrow \infty} \left[\int_{-B}^0 f(t) dt + \int_0^B f(t) dt \right]$$

$$\Rightarrow \lim_{B \rightarrow \infty} \int_0^B f(t) dt = \lim_{B \rightarrow \infty} \int_0^B a\lambda_f e^{-\lambda_f t} + (1-a)\lambda_s e^{-\lambda_s t} dt$$

$$= \lim_{B \rightarrow \infty} \left[a e^{-\lambda_f t} + (1-a) e^{-\lambda_s t} \right]_0^B - 1$$

$$= \lim_{B \rightarrow \infty} \left[a e^{-\lambda_f B} + (1-a) e^{-\lambda_s B} - (a + 1-a) \right] = (-1) = (-1)(-1) = 1$$

$$MGF = E[e^{st}] = \int_{-\infty}^{\infty} e^{st} f(t) dt = \int_0^{\infty} a\lambda_f e^{-\lambda_f t} e^{st} + (1-a)\lambda_s e^{-\lambda_s t} e^{st} dt$$

$$= \lim_{B \rightarrow \infty} \int_0^B a\lambda_f e^{(s-\lambda_f)t} + (1-a)\lambda_s e^{(s-\lambda_s)t} dt = \lim_{B \rightarrow \infty} \left[\frac{a\lambda_f}{s-\lambda_f} e^{(s-\lambda_f)t} + \frac{(1-a)\lambda_s}{s-\lambda_s} e^{(s-\lambda_s)t} \right]_0^B$$

comparable for $s < \lambda_f, s < \lambda_s$

$$= \frac{-a\lambda_f}{s-\lambda_f} + \frac{(1-a)\lambda_s}{s-\lambda_s} = \frac{a\lambda_f}{\lambda_f - s} + \frac{(1-a)\lambda_s}{\lambda_s - s}$$

$$E[T] = \frac{dV_f(0)}{ds} = \frac{a\lambda_f}{(\lambda_f)^2} + \frac{(1-a)\lambda_s}{(\lambda_s)^2} = \frac{a}{\lambda_f} + \frac{1-a}{\lambda_s}$$

$$E[T^2] = \frac{d^2 V_f(0)}{ds^2} = \frac{2a\lambda_f \cdot \lambda_f}{(\lambda_f)^4} + \frac{2(1-a)\lambda_s \cdot \lambda_s}{(\lambda_s)^4} = \frac{2a}{\lambda_f^2} + \frac{2(1-a)}{\lambda_s^2}$$

$$a = .9 \quad \lambda_f = 10005^{-1} \quad \lambda_s = 105^{-1}$$

$$2.a) \quad E[T] = \frac{a}{\lambda_f} + \frac{1-a}{\lambda_s} \quad E[T^2] = \frac{2a}{\lambda_f^2} + \frac{2(1-a)}{\lambda_s^2}$$

$$\text{Var}(T) = E[T^2] - (E[T])^2 = \frac{2a}{\lambda_f^2} + \frac{2(1-a)}{\lambda_s^2} - \left(\frac{a}{\lambda_f} + \frac{1-a}{\lambda_s} \right)^2$$

$$\text{Var}(T) = \frac{2a}{\lambda_f^2} + \frac{2(1-a)}{\lambda_s^2} - \frac{a^2}{\lambda_f^2} - \frac{(1-a)^2}{\lambda_s^2} - \frac{2a(1-a)}{\lambda_f \lambda_s} \approx .00133299$$

$$E[T] = \frac{a}{\lambda_f} + \frac{1-a}{\lambda_s} \approx .0109$$

$$CV = \frac{\sqrt{\text{Var}(T)}}{E[T]} \approx \frac{\sqrt{.00133299}}{.0109} \approx 3.98105 \approx CV$$

Exponential distribution $E[X] = \frac{1}{\lambda} \quad \text{Var}(X) = \frac{1}{\lambda^2}$

$$CV = \frac{\sqrt{\text{Var}(X)}}{E[X]} = \frac{\frac{1}{\lambda}}{\frac{1}{\lambda}} = 1$$

CV of the Ion Channel is about 4 times higher, which means that the Ion Channel are less clustered to the expectation of mean

b) Bernoulli draw Gives whether λ_s or λ_f term need inverse of exponential

$$F(x) = 1 - e^{-\lambda x} \quad F(x) = \int_{-\infty}^x f(x) dx = \int_0^x \lambda e^{-\lambda x} dx = \left[-e^{-\lambda x} \right]_0^x = 1 - e^{-\lambda x}$$

$$x = 1 - e^{-\lambda u} \quad 1 - x = e^{-\lambda u} \Rightarrow \log(1-x) = -\lambda u \quad u = \frac{\log(1-x)}{-\lambda}$$

$$F^{-1}(x) = \frac{-\log(1-x)}{\lambda}$$

Algorithm

$$U \sim \text{uniform}(0,1) = \text{rand}(1)$$

$$V \sim \text{uniform}(0,1) = \text{rand}(1)$$

if $U < a$

$$F^{-1}(V) = \frac{-\log(1-V)}{\lambda_f}$$

else

$$F^{-1}(V) = \frac{-\log(1-V)}{\lambda_s}$$

c) Empirical Mean $M = .0109$ direct match to analytical mean

$$\text{Empirical } P(T \geq 50\text{ms}) \approx .0602$$

$$P(T \leq t) = F(t) = 1 - ae^{-\lambda_f t} - (1-a)e^{-\lambda_s t}$$

$$\text{let } P(T \geq 50\text{ms}) = 1 - P(T \leq 50\text{ms}) = ae^{-\lambda_f(50)} + (1-a)e^{-\lambda_s(50)} \approx .06075$$

Empirical was 5×10^{-4} off, still very accurate

5. $C \sim N^w$ $w \approx 0.566$

$$C(N+1) = \begin{cases} C(N)+1, & \text{w/ Probability } w(C(N)/N \\ C(N), & \text{otherwise} \end{cases}$$

a) $h(z) = 2(1-z)$ $0 < z < 1$

if h is a density $\int_{-\infty}^{\infty} h(z) dz = 1$

$$\Rightarrow \int_{-\infty}^{\infty} 2(1-z) dz = \int_0^1 2(1-z) dz = [2z - z^2]_0^1 = 2 - 1 = 1$$

$$E[z] = \int_{-\infty}^{\infty} z h(z) dz = \int_{-\infty}^{\infty} 2z - 2z^2 dz = \int_0^1 2z - 2z^2 dz = [z - \frac{2}{3}z^3]_0^1 = 1 - \frac{2}{3} = \frac{1}{3}$$

$$E[z] = \frac{1}{3}$$

$$E[z^2] = \int_{-\infty}^{\infty} z^2 h(z) dz = \int_{-\infty}^{\infty} 2z^2 - 2z^3 dz = \int_0^1 2z^2 - 2z^3 dz = [\frac{2}{3}z^3 - \frac{1}{2}z^4]_0^1 = \frac{2}{3} - \frac{1}{2} = \frac{1}{6}$$

$$\text{Var}(z) = E[z^2] - E[z]^2 = \frac{1}{6} - \frac{1}{9} = \frac{9}{54} - \frac{6}{54} = \frac{3}{54} = \frac{1}{18} = \text{Var}(z)$$

$$CV = \frac{\sigma}{E[z]} = \frac{\sqrt{\text{Var}(z)}}{E[z]} = \frac{\sqrt{\frac{1}{18}}}{\frac{1}{3}} \approx 0.7071 = \frac{1}{\sqrt{2}} = CV$$

$$= \frac{3}{3\sqrt{2}} = \frac{1}{\sqrt{2}}$$

5. b) $h(z) = 2(1-z)$

$$H(z) = \int_{-\infty}^z h(z) dz = \int_0^z h(z) dz = \left[2z - z^2 \right]_0^z = 2z - z^2 \text{ for } 0 \leq z \leq 1$$

let $X \sim \text{Uniform}(0,1)$

$$z = 2x - x^2 = x(2-x)$$

$$\Rightarrow 1-z = x^2 - 2x + 1$$

$$\Rightarrow 1-z = (x-1)^2$$

$$\Rightarrow x = \pm \sqrt{1-z} + 1 \quad \text{since } 0 \leq x \leq 1 \text{ and } 0 \leq z \leq 1 \text{ take -}$$

$$\Rightarrow x = 1 - \sqrt{1-z}$$

$$F^{-1}(z) = 1 - \sqrt{1-z}$$

PSudo code

for $i=1:n$

$$z_i = \text{rand}(1)$$

$$F^{-1}(z_i) = 1 - \sqrt{1-z_i}$$

end

calculated mean $M = .3336 \approx \frac{1}{3}$ Nearly direct match to analytical mean